

International Marketing Analytics

Logistic regression across countries

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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Today

Odds | Churn logit | Countries | Using the model | Python | In class

Theory, Parts 1 to 4

- Odds, odds ratios, the S-curve
- Reading the churn logit
- Comparing countries
- Predict, rank, target

Practice, Parts 5 and 6

- Code ladder: one line to the full model
- Online quiz and its answers
- Coding exercise: churn in four countries
- The same tasks with Posit Assistant

Question of the day: who will cancel, and does it differ by country?

Our example: a mobile operator in four countries

The problem: a mobile operator in four countries is losing customers.

The question: who will cancel in the next 12 months, and what can we do?

4,000 customers in **Germany**	**1,500** customers in **Austria**	**3,000** customers in **Spain**	**2,500** customers in **Poland**

Churn = 1: the customer cancelled within the next 12 months; 0: the customer stayed.

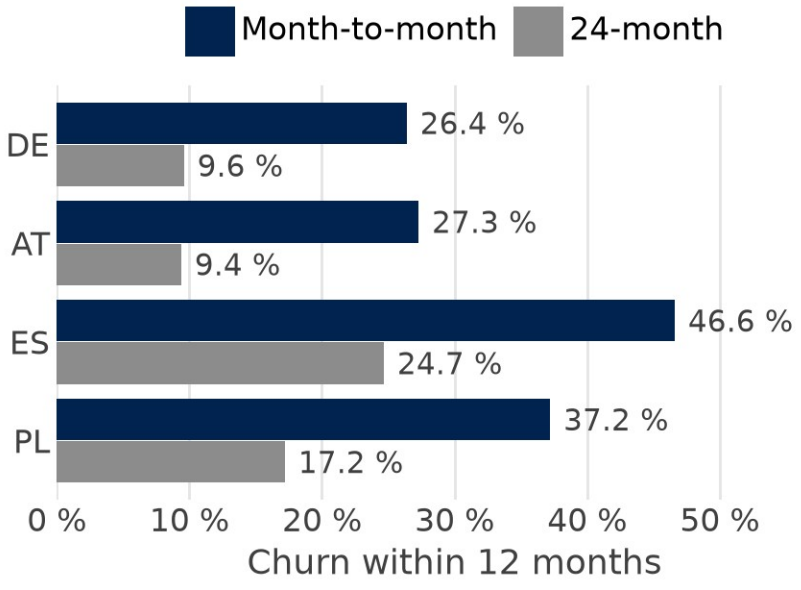
Simulated data (telco_churn.csv), built for this course

The data: one row per customer

Column	Meaning	Customer 47
customer_id	customer number	47
country	DE, AT, ES or PL	DE
age	age in years	20
tenure_months	months with the operator	10
monthly	1 = month-to-month contract, 0 = 24-month	1
monthly_fee	fee in EUR per month	27.22
price_increase	1 = price rise in the last six months	0
complaints	complaint calls in the last six months	1
satisfaction	survey score, 1 to 5	2
retention_offer	1 = got the retention offer (random pilot)	0
churn	1 = cancelled within the next 12 months	1

File: `data/churn/telco_churn.csv` simulated, 11,000 customers

First look: churn by country and contract



1 in 4 **month-to-month** customers in Germany churn (26.4 %)

1 in 10 **24-month** customers in Germany churn (9.6 %)

The contract matters everywhere – how much? That needs a model for a yes/no outcome.

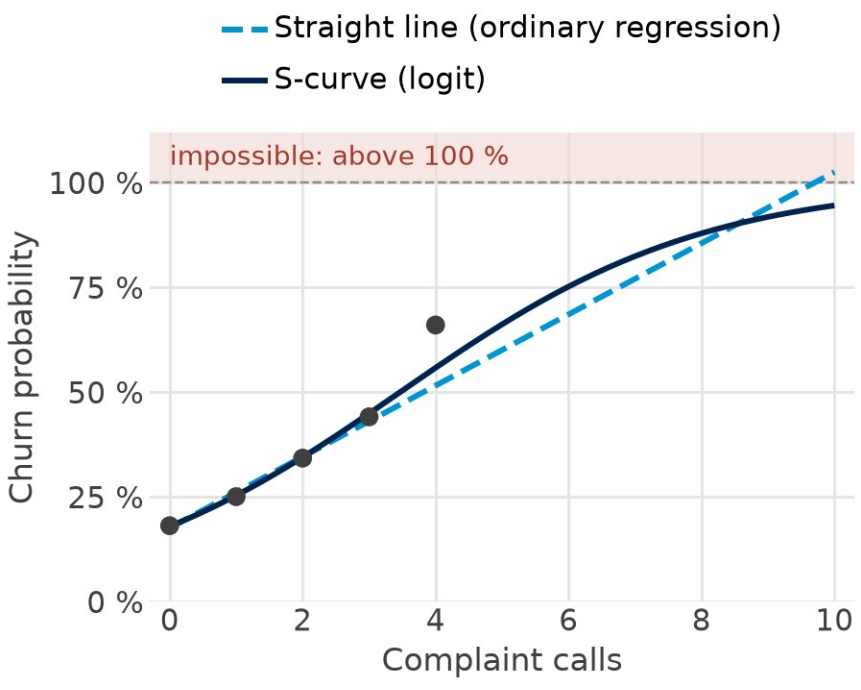
Simulated data (telco_churn.csv)

From churn rates to odds



Part 1 of 6 · odds, odds ratios, the S-curve

Why not ordinary regression for churn?



Straight line: easy to read, but no limits: 103 % churn at 10 complaint calls.

S-curve (logit): always between 0 and 100 %; at 10 calls 95 %.

The price of the S-curve: a coefficient no longer says directly how much the probability changes.

Simulated data (telco_churn.csv); dots: churn rate at 0 to 4 calls

Worked example: 100 + 100 German customers

German customers	Churners	Customers	Probability p	Odds = p / (1 − p)
Month-to-month	25	100	0.25	0.25 / 0.75 = 0.33
24-month	10	100	0.10	0.10 / 0.90 = 0.11

Formula

```
p    = churners / customers
odds = p / (1 − p)
     = churners / stayers
```

Odds 0.33 = 1/3: one churner per three stayers

Odds 0.11 = 1/9: one churner per nine stayers

Odds = churners per stayer: 0 means nobody churns, 1 means 50:50.

Invented round numbers, close to the simulated data (Germany: 26 % vs 10 %)

Odds ratio and log-odds: the same example

3.0

Odds ratio = 0.33 / 0.11
a month-to-month contract triples the odds of churn

1.10

Log-odds = ln odds: ln(1/3) − ln(1/9) = −1.10 − (−2.20)
= ln 3: the logit coefficient of month-to-month

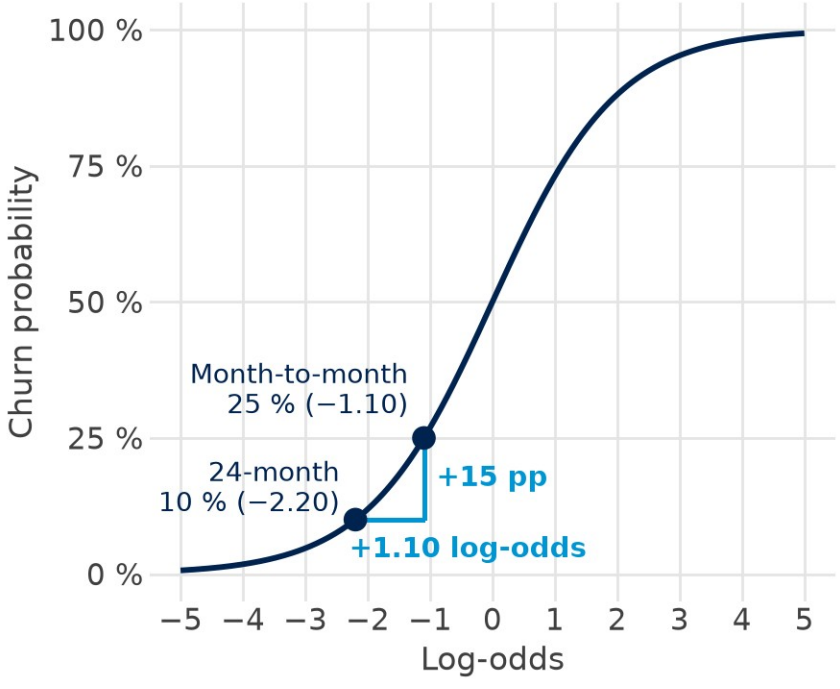
0.25

Back to probability: p = odds / (1 + odds)
0.33 / 1.33 = 0.25

Not “3 times as likely”: churn rises from 10 % to 25 %: 2.5 times, +15 percentage points (pp).

Invented round numbers, close to the simulated data (slide 8)

One model, three scales



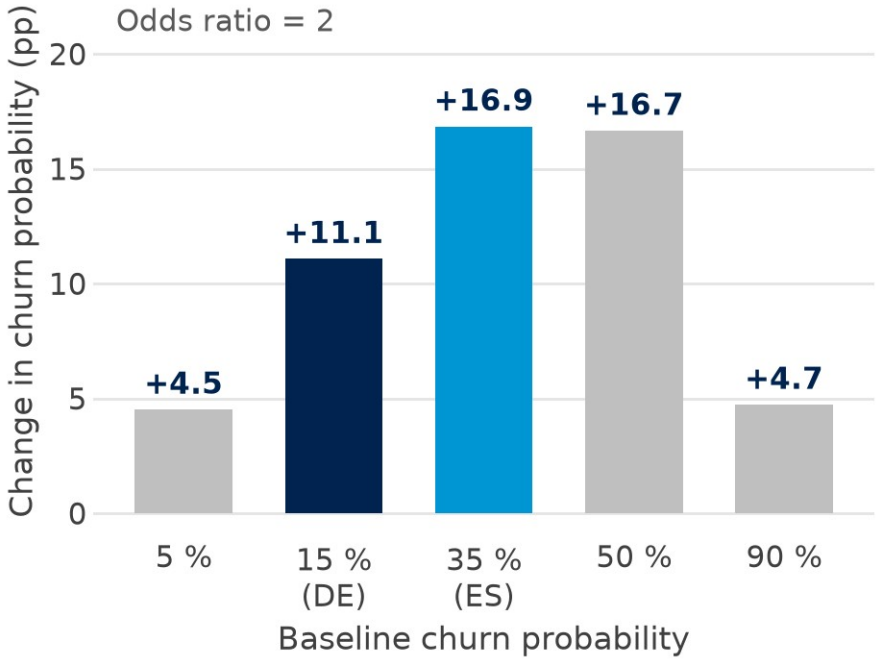
Log-odds: −2.20 → −1.10, +1.10. Effects add up; the output table shows these.

Odds: 0.11 → 0.33, × 3.0. Effects multiply: the odds ratio.

Probability: 10 % → 25 %, +15 pp. What a manager wants; the change depends on the start.

Worked example of slide 8; curve: p = 1 / (1 + e^(−log-odds))

An odds ratio of 2 is not “twice as likely”



× 1.9 **5 % → 9.5 %**
nearly twice as likely

× 1.5 **35 % → 51.9 %**
1.5 times as likely, +16.9 pp

“Twice as likely” only works for rare outcomes, below about 10 %. Churn of 15 to 35 % is not rare.

Own calculation: $p_1 = OR \cdot p_0 / (1 - p_0 + OR \cdot p_0)$, with p_0 the starting churn rate

Your turn: Spanish customers, with a calculator

Spanish customers	Churners	Customers	p	Odds	Log-odds
Month-to-month	45	100	?	?	?
24-month	25	100	?	?	?

1 p and odds of both groups

2 the odds ratio, and the difference in log-odds

3 compare with Germany (odds ratio 3.0, +15 pp): odds ratio and gap in percentage points

Pairs, 5 minutes: where does the contract matter more, Germany or Spain?

Invented round numbers, close to the simulated data (Spain: 47 % vs 25 %)

Answers: in class

The churn logit

Part 2 of 6 · reading the output, three numbers to report

The churn logit: reading the table

	Term	Coefficient	Odds ratio
1	**Month-to-month**	0.962	2.62
	Price rise	0.602	1.83
2	**Complaint calls (each)**	0.452	1.57
	ln tenure (months)	−0.240	0.79
	Austria (vs Germany)	−0.010	0.99
3	**Spain (vs Germany)**	1.057	2.88
	Poland (vs Germany)	0.618	1.86

1. **Contract, OR 2.62:** month-to-month multiplies the odds of churn by 2.6

2. **Complaints, OR 1.57:** each call raises the odds by 57 %

3. **Spain, OR 2.88:** higher churn odds than Germany at equal drivers

Above 1 raises the odds, below 1 lowers them; percentage points come next.

Simulated data (telco_churn.csv); one logit for all countries, Germany = reference

Three numbers to report

Odds ratio

Multiplies the odds of churn

Use: inside one model

2.62

month-to-month

Marginal effect

Change in churn probability, in pp

Use: compare countries

+15.6 pp

month-to-month, on average

Prediction

Churn probability of one profile

Use: talk to managers

DE 38.1 %

95 % CI 34.1–42.2 %

ES 58.0 %

95 % CI 54.0–62.0 %

Report percentage points and probabilities; keep odds ratios in the table.

Simulated data (telco_churn.csv); prediction: month-to-month, price rise, one complaint, 12 months

Comparing countries

Part 3 of 6 · coefficients, probabilities, survey scores

The puzzle: where does the contract matter more?

One logit per country:
month-to-month coefficient

Germany 1.19
Spain 0.81

Does the contract matter more in Germany?
Show of hands: yes or no?

What else differs?
Churn: 15.4 % in Germany,
34.7 % in Spain

What did slide 12 show?
Odds ratios and pp can rank countries differently

Simulated data (telco_churn.csv); one logit per country, four drivers

Same effect, different coefficients

Country	True effect	Noise scale	Coefficient
Germany	1.3	1.0	1.19
Austria	1.3	1.0	1.02
Spain	1.3	1.5	0.81
Poland	1.3	1.2	0.90

Coefficients from one logit per country

Rule of thumb

```
coefficient ≈
  true effect / noise

Germany  1.3 / 1.0 = 1.30
Spain    1.3 / 1.5 ≈ 0.87
```

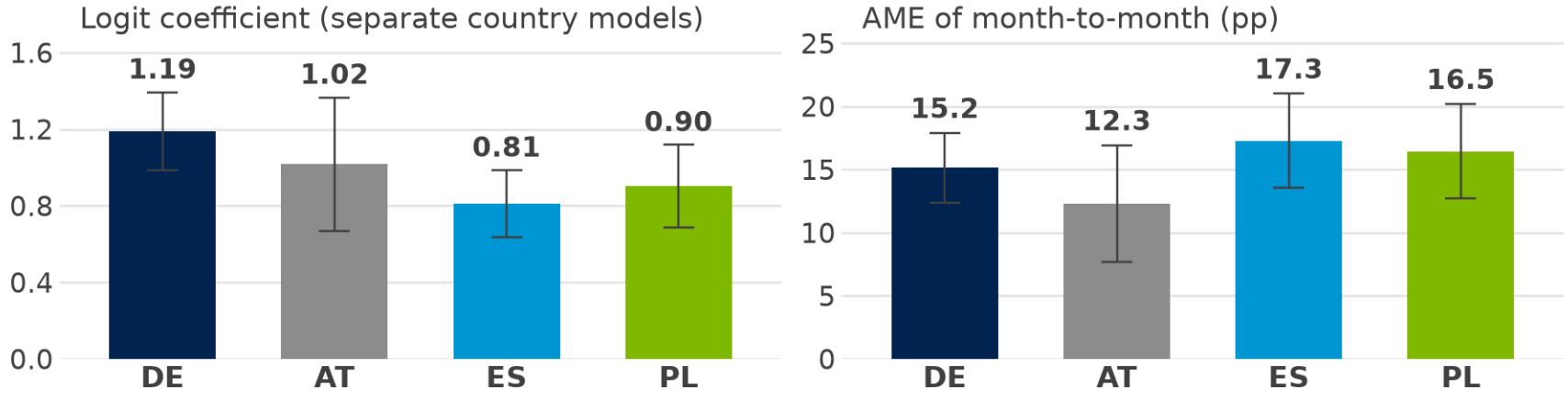
Noise scale: how much churn depends on things the model does not see

Naive test: 1.19 vs 0.81 seems a real gap ($z = 2.8$)

Smaller coefficient, same effect: more noise in Spain shrinks the coefficient.

Simulated data (telco_churn.csv); true effect and noise set in the simulation

On the probability scale the gap disappears



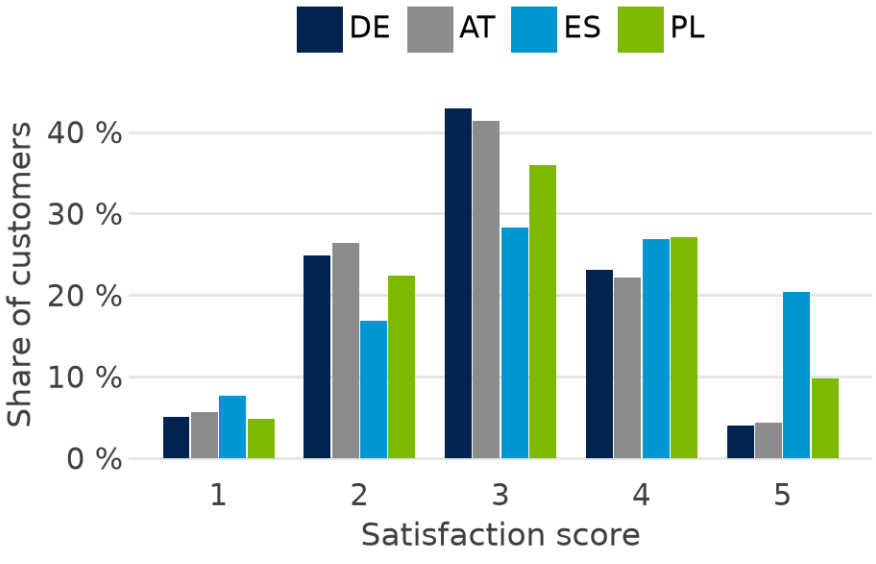
Lines: 95 % CIs. Gap Spain − Germany 2.1 pp: within sampling error

Noise: more of it shrinks Spain's coefficient (1.5 vs 1.0)

Baseline: at 34.7 % churn the S-curve is steeper than at 15.4 %

Simulated data (telco_churn.csv); one logit per country

Survey scores differ by country too



20.3 % **give a 5 in Spain** (Germany 4.0 %)

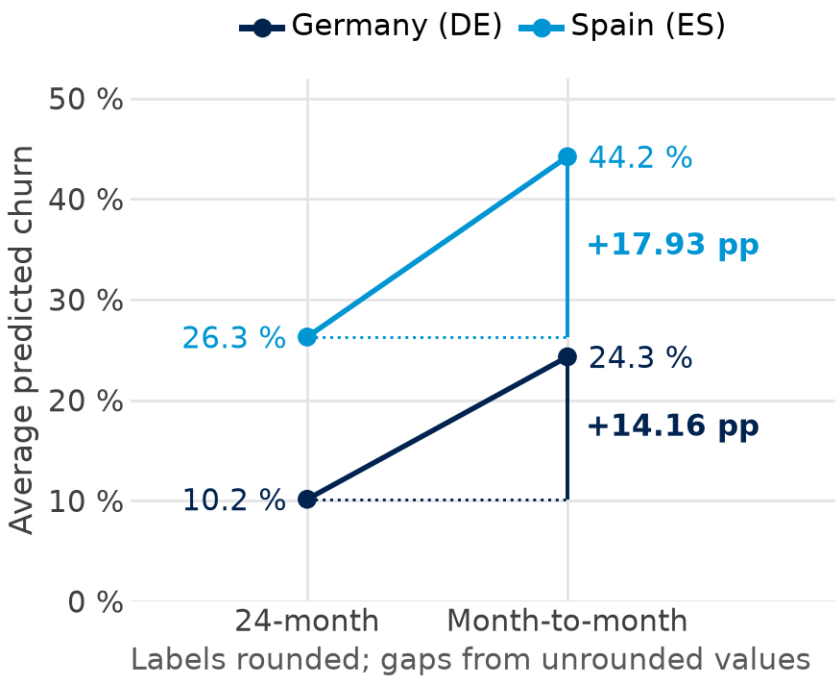
3.35 **mean score in Spain** (Germany 2.96)

Equal **true satisfaction** in every country (simulation)

Same satisfaction, different scores: standardise within each country before you pool.

Simulated data (telco_churn.csv); answer styles set in the simulation

Interactions: read them in percentage points



−0.245 **Product term** in log-odds: “weaker in Spain”

+3.8 pp **In pp:** 17.93 (Spain) minus 14.16 (Germany): “stronger”

Read interactions in pp. The country models give DE 15.2 pp and ES 17.3 pp: their 95 % intervals overlap.

Simulated data (telco_churn.csv); one logit with contract × country

Using a churn model



Part 4 of 6 · predict, rank, target

Who will churn? The threshold and the confusion matrix

Fit on 70 % → Test on 30 % → **Pick a threshold** → Call the flagged

Threshold 0.5	Churned	Stayed
Flagged	175	106
Not flagged	619	2,384

77.9 % **accuracy;** “nobody churns” scores 75.8 %

22 % **of churners caught** at 0.5; 56 % at 0.3

Flagged: predicted churn probability ≥ 0.5

Accuracy misleads when most customers stay: count the churners you catch; lower the threshold if missing one costs more than a wasted offer.

Simulated data (telco_churn.csv), 3,284 test customers

The math behind the confusion matrix

At 0.5	Churned	Stayed
Flagged	TP 175	FP 106
Not flagged	FN 619	TN 2,384

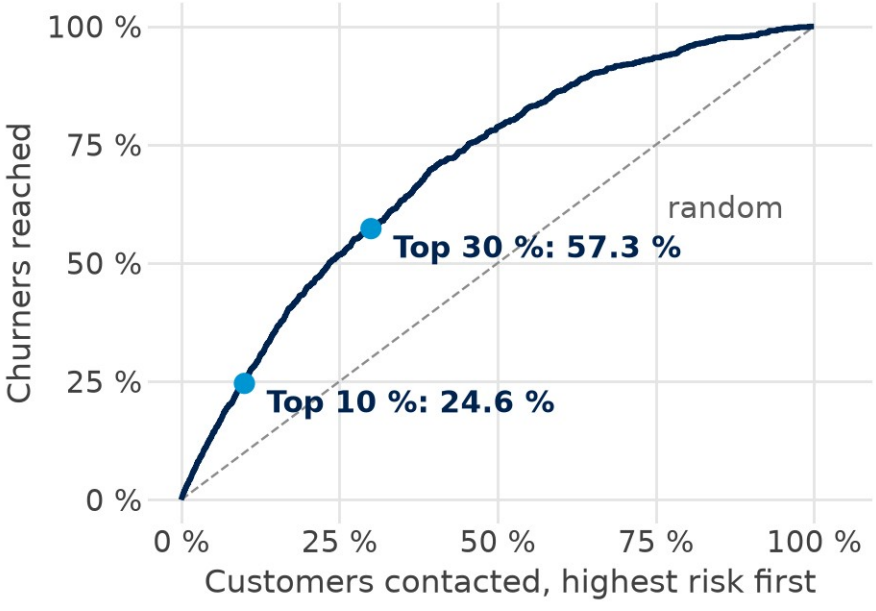
Cell	Meaning
TP	true positive: flagged, churned
FP	false positive: flagged, stayed
FN	false negative: a missed churner
TN	true negative: not flagged, stayed

Measure	Formula	Threshold 0.5, by hand	At 0.3
Accuracy	(TP + TN) / all	(175 + 2,384) / 3,284 = 77.9 %	74.2 %
Recall	TP / (TP + FN)	175 / 794 = 22.0 %	56.2 %
Precision	TP / (TP + FP)	175 / 281 = 62.3 %	47.2 %
Specificity	TN / (TN + FP)	2,384 / 2,490 = 95.7 %	80.0 %

A lower threshold trades precision for recall: more churners caught, more wasted offers.

Simulated data (telco_churn.csv), 3,284 test customers; TP, FP, FN, TN as on slide 24

Ranking customers: the top-decile lift



2.46 — **top-decile lift:** 59.5 % churn in the top 10 %, 24.2 % overall

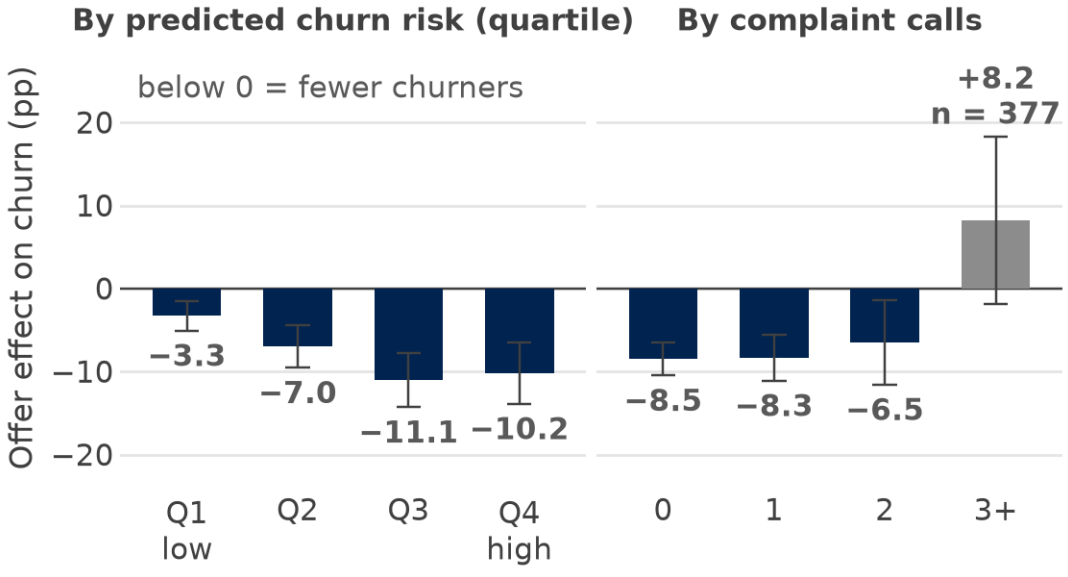
24.6 % — **of all churners** are in the top 10 %

57.3 % — **of all churners** are in the top 30 %

Fixed budget? Call from the top of the ranked list.

Sources: Neslin et al. (2006); simulated data (telco_churn.csv), test customers

Whom to call? Risk is not uplift



Pilot: a random half got a retention offer; uplift = churn with minus without it

Highest risk ≠ largest effect: Q4 −10.2 pp, Q3 −11.1 pp

No response from 3 complaint calls on (377 customers; the interval includes 0)

Call the customers the offer moves, not the riskiest: it takes a random control group.

Sources: Ascarza (2018); simulated data (telco_churn.csv), offer pilot

The analyses in Python

Part 5 of 6 · a code ladder, one step at a time

The packages and functions

Task	Package	Function
Load and look	polars	pl.read_csv(); group_by().agg(); filter()
Fit the logit	statsmodels	smf.logit(formula, data=df).fit()
Read the output	statsmodels	m.summary()
Odds ratios	numpy	np.exp(m.params)
Percentage points (AME)	statsmodels	m.get_margeff(at="overall", dummy=True)
Predict a customer	statsmodels	m.predict(new)
Chart (exercise)	plotnine	ggplot() + geom_col()

Step 1: load and look

logit_churn.qmd · Step 1

```
import polars as pl

# load the data and show the first rows
churn = pl.read_csv("../../data/churn/telco_churn.csv")
print(churn.head())

# churn rate by country
rates = churn.group_by("country").agg(pl.col("churn").mean())
rates.sort("country")
```

country	churn
AT	0.146
DE	0.154
ES	0.347
PL	0.285

Look first: Spain and Poland churn about twice as often as Germany and Austria.

Step 2: one variable, one line

logit_churn.qmd · Step 2

```
import statsmodels.formula.api as smf

# statsmodels needs a pandas table
df = churn.to_pandas()

# fit a logit with one variable: the contract
m1 = smf.logit("churn ~ monthly", data=df).fit()
m1.summary()
```

m1.summary()	coef	P>|z|
Intercept	−1.759	0.000
monthly	1.176	0.000

Read sign and p: month-to-month customers churn more, and it is not chance.

Step 3: odds ratios

logit_churn.qmd · Step 3

```
import numpy as np

# odds ratios from the coefficients
np.exp(m1.params)
```

np.exp(m1.params)	odds ratio
Intercept	0.17
monthly	3.24

3.24 times the odds: of churning for month-to-month customers, not 3.24 times the probability.

Step 4: add a second variable

logit_churn.qmd · Step 4

```
# add a second variable: complaint calls
m2 = smf.logit("churn ~ monthly + complaints",
               data=df).fit()
m2.summary()
```

coef	m1	m2
monthly	1.176	1.231
complaints	–	0.483

Held constant: the contract still raises churn among customers with the same number of complaint calls.

Step 5: all drivers and the countries

logit_churn.qmd · Step 5

```
# add price rise, tenure and the countries,
# with Germany as the reference country
m3 = smf.logit("churn ~ monthly + price_increase"
               " + complaints + np.log(tenure_months)"
               " + C(country, Treatment('DE'))",
               data=df).fit()
m3.summary()
```

m3.summary()	coef	P>|z|
monthly	0.962	0.000
price_increase	0.602	0.000
complaints	0.452	0.000
np.log(tenure_months)	−0.240	0.000
country AT	−0.010	0.907
country ES	1.057	0.000
country PL	0.618	0.000

Germany is the reference: Spain and Poland churn more with the same drivers, Austria does not differ.

Step 6: percentage points: the AME

logit_churn.qmd · Step 6

```
# average marginal effects: the change in the
# churn probability (0.10 = 10 percentage points)
m3.get_margeff(at="overall", dummy=True).summary()
```

get_margeff	dy/dx	interval
monthly	0.156	0.138 to 0.174
price_increase	0.098	0.081 to 0.115
complaints	0.070	0.062 to 0.078
country ES	0.179	0.158 to 0.201

0.156 = 15.6 percentage points: a month-to-month contract raises churn by that much on average.

Step 7: predict a customer

logit_churn.qmd · Step 7

```
# two new customers: month-to-month, a price rise,
# one complaint call, 12 months; Germany and Spain
new = pl.DataFrame({
    "country": ["DE", "ES"],
    "monthly": 1, "price_increase": 1,
    "complaints": 1, "tenure_months": 12})
m3.predict(new.to_pandas())
```

m3.predict(new)	churn
Germany	0.351
Spain	0.609

Same customer, other country: 35.1 % churn in Germany, 60.9 % in Spain.

Step 8: one country at a time

logit_churn.qmd · Step 8 · Germany

```
# the same drivers, Germany only
de = churn.filter(pl.col("country") == "DE")
m_de = smf.logit(
    "churn ~ monthly + price_increase"
    " + complaints + np.log(tenure_months)",
    data=de.to_pandas()).fit()
m_de.get_margeff(at="overall",
                 dummy=True).summary()
```

logit_churn.qmd · Step 8 · Spain

```
# the same drivers, Spain only
es = churn.filter(pl.col("country") == "ES")
m_es = smf.logit(
    "churn ~ monthly + price_increase"
    " + complaints + np.log(tenure_months)",
    data=es.to_pandas()).fit()
m_es.get_margeff(at="overall",
                 dummy=True).summary()
```

monthly	coef	AME	95 % interval
Germany	1.189	0.152	0.124 to 0.179
Spain	0.810	0.173	0.135 to 0.210

Compare in pp: the AME intervals overlap, so the gap is within chance; the coefficients rank the other way.

Step 9: how good is the model?

logit_churn.qmd · Step 9

```
# predicted churn probability of every customer
risk = m3.predict(df).to_numpy()
# flag the customers with a risk of 0.5 or more
scored = churn.with_columns(flagged=risk >= 0.5)
# churn rate and number of customers per group
groups = scored.group_by("flagged").agg(
    pl.col("churn").mean(), pl.len())
groups.sort("flagged")
```

flagged	churn	customers
false	0.205	10,116
true	0.579	884

Flagged customers churn more: 57.9 % against 20.5 % of the others.

In-class exercise

Part 6 of 6 · quiz, coding exercise, Posit Assistant

Online quiz: logit and churn in five questions

1 In Austria, 20 of 100 month-to-month customers churn. The odds of churn are …
A 0.20 B 0.25 C 4.0

2 An odds ratio of 2 at a baseline churn rate of 20 %. The new churn probability is …
A 33 % B 40 % C 22 %

3 Month-to-month coefficient: 1.19 in Germany, 0.81 in Spain. So …
A bigger effect in Germany B not clear yet: compare in pp C bigger effect in Spain

4 Product term month-to-month × Spain: −0.245. In pp, the effect in Spain …
A is smaller than in Germany B can still be larger than in Germany C is zero

5 Who should get the retention offer?
A the highest predicted risk B those whose churn it lowers most C most complaint calls

3 minutes alone, then compare with your neighbour.

Answers: in class

Your turn: churn across four countries, in Python

Pairs, code ladder: 35 minutes	Plenary, answers: 5 minutes	Posit Assistant: 10 minutes

1. **Fit:** m3 of Step 5: odds ratio, AME and predictions for Austria and Poland (Steps 3, 6, 7)
2. **Compare:** Step 8 for Austria: coefficient vs AME, next to Spain; two memo sentences
3. **Evaluate:** Step 9 at a threshold of 0.3 instead of 0.5: who is flagged, who churns?
4. **Target:** churn with and without the offer by complaint calls; who gets the offer? (code given)

Deliverable: one slide per pair with a plot and two sentences for the retention manager.

File: `sessions/01-foundations/churn_exercise.qmd` (code to copy: logit_churn.qmd)

Then: the same tasks with Posit Assistant

How: code first, then open Posit Assistant, type @churn_exercise.qmd, one task per prompt

Prompt for task 1

```
@churn_exercise.qmd Task 1. Fit m3 as
in Step 5. Explain the odds ratio and
the AME of monthly, and predict the
customer of Step 7 in AT and PL.
```

Prompt for task 2

```
Task 2. Fit the model of Step 8 for
Austria. Why do the coefficients and
the AMEs rank Austria and Spain
differently?
```

Prompt for task 3

```
Task 3. Run Step 9 with a threshold
of 0.3 instead of 0.5. How many are
flagged, and how many of them churn?
Which threshold suits a campaign?
```

Prompt for task 4

```
Task 4. Show churn with and without
the offer by complaint calls, in a
table and a chart. Who should get the
offer, and why not the riskiest?
```

Check every number against your own code: you own the result.

Open churn data for your project

IBM Telco Churn

7,043 customers, USA (fictional)

Churn: Left in the last month

Use: Learn the mechanics

Licence: Credit IBM

Bank churn

Churn_Modelling: 10,000 customers in France, Germany, Spain

Churn: Closed the account

Use: A country effect

Licence: Unclear: link only

Online Retail II

1.07 m transactions, 43 countries

Churn: You define it

Use: Real data, own label

Licence: CC BY 4.0

The licence decides: commit only data you may share; link the rest.

Guide: `sessions/01-foundations/churn_readings_and_data.md` (links, polars code, traps)

Sources: dataset checks, Oct 2026 (churn_readings_and_data.md)

Key takeaways

1. **Odds are not probabilities:** odds = churners per stayer; OR 3 is not “3 times as likely”
2. **Report three numbers:** odds ratio, AME in percentage points, predicted probability with CI
3. **Compare countries in pp:** not in coefficients, which carry each country's noise
4. **Interactions and survey scores:** read interactions in pp; check scores before pooling
5. **Rank by risk (lift), target by uplift:** the offer goes to those it changes

Next: run sessions/01-foundations/logit_churn.qmd, the code ladder, step by step.

References

Logit methods

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Ascarza, E. (2018). Retention futility: targeting high-risk customers might be ineffective. JMR, 55(1), 80–98.

Full list with DOIs: `sessions/01-foundations/churn_readings_and_data.md`

Thank you. Questions and discussion

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